SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 40%

Code of Conduct:

I LAKHANA PRAVEEN 192426049 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L. Praveen

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks	Q5&M
Problem Understanding	20%	1	2	2	2	2
Method Selection	20%	1	2	1	1	2
Solution Process	25%	2	1	2	2	2
Accuracy of Calculation	20%	1	1	1	1	1
Interpretation of Results	15%	1	1	1	1	1

Total Marks 35 /40

Signature of the Faculty

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Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks **/40** ***Signature of the Faculty***

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical,	Mostly	Disorganized	No clear

		systematic steps with correct approach throughout	correct steps with minor mistakes	steps; multiple errors	process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

SOLUTIONS

Question 1:

Problem Summary & Base Allocation:

- Given Base Block: 192.168.20.0/24
- Total IP addresses in /24 block: 256 addresses (192.168.20.0 to 192.168.20.255)

Subnet Calculation & Analysis:

- **Lab A (/26 subnet):** Subnet mask is 255.255.255.192 (/26).

  Block size = 2^(32-26) = 64 addresses.
  - Network Address: 192.168.20.0/26
  - Broadcast Address: 192.168.20.63
  - Valid Host IP Range: 192.168.20.1 to 192.168.20.62
  - Total Usable Hosts: 2^6 - 2 = 62 usable addresses.

- **Lab B (/27 subnet):** Allocated next available address space starting at 192.168.20.64. Subnet mask is 255.255.255.224 (/27).

  Block size = 2^(32-27) = 32 addresses.
  - Network Address: 192.168.20.64/27
  - Broadcast Address: 192.168.20.95
  - Valid Host IP Range: 192.168.20.65 to 192.168.20.94
  - Total Usable Hosts: 2^5 - 2 = 30 usable addresses.

- **Lab C (/28 subnet):** Allocated next available address space starting at 192.168.20.96. Subnet mask is 255.255.255.240 (/28).

  Block size = 2^(32-28) = 16 addresses.
  - Network Address: 192.168.20.96/28
  - Broadcast Address: 192.168.20.111
  - Valid Host IP Range: 192.168.20.97 to 192.168.20.110
  - Total Usable Hosts: 2^4 - 2 = 14 usable addresses.

Summary Table for Question 1:

Lab	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
Lab A	/26 (255.255.255.192)	192.168.20.0	192.168.20.1 – 192.168.20.62	192.168.20.63	62
Lab B	/27 (255.255.255.224)	192.168.20.64	192.168.20.65 – 192.168.20.94	192.168.20.95	30
Lab C	/28 (255.255.255.240)	192.168.20.96	192.168.20.97 – 192.168.20.110	192.168.20.111	14

Question 2:

Problem Analysis & Given Data:

- System 1: IP = 192.168.1.130, Mask = 255.255.255.192 (/26)
- System 2: IP = 192.168.1.190, Mask = 255.255.255.192 (/26)

Subnet Range & Network Calculations:

- Subnet Mask Analysis: 255.255.255.192 = 11111111.11111111.11111111.11000000. Block size = 256 - 192 = 64. Subnet increments: 0, 64, 128, 192.
- System 1 Analysis (192.168.1.130): 130 falls in the range 128 to 191 (Subnet: 192.168.1.128/26).
  - Network Address: 192.168.1.128
  - Broadcast Address: 192.168.1.191
  - Subnet Range: 192.168.1.128 to 192.168.1.191
  - Valid Host IP Range: 192.168.1.129 to 192.168.1.190
  - Verification: 192.168.1.130 IS within its valid host IP range.
  - Usable Hosts: 2^6 - 2 = 62 usable addresses.
- System 2 Analysis (192.168.1.190): 190 also falls in the range 128 to 191 (Subnet: 192.168.1.128/26).
  - Network Address: 192.168.1.128
  - Broadcast Address: 192.168.1.191
  - Subnet Range: 192.168.1.128 to 192.168.1.191
  - Valid Host IP Range: 192.168.1.129 to 192.168.1.190
  - Verification: 192.168.1.190 IS within its valid host IP range (last usable host IP).
  - Usable Hosts: 2^6 - 2 = 62 usable addresses.

Conclusion :

- **Note:** Both System 1 and System 2 belong to the SAME subnet (192.168.1.128/26) and both IP addresses fall within the valid host range.

Parameter	System 1 (192.168.1.130)	System 2 (192.168.1.190)
Subnet Mask	255.255.255.192 (/26)	255.255.255.192 (/26)
Network Address	192.168.1.128	192.168.1.128
Broadcast Address	192.168.1.191	192.168.1.191
Subnet Range	192.168.1.128 – 192.168.1.191	192.168.1.128 – 192.168.1.191
Valid Host IP Range	192.168.1.129 – 192.168.1.190	192.168.1.129 – 192.168.1.190
IP Within Range?	Yes (Valid Host)	Yes (Valid Host - Last Usable IP)
Usable Host Count	62	62

Question 3:

Problem Summary & Base Allocation:

- Given Base Block: 172.16.0.0/16

Departmental Subnet Allocation:

- **1. HR Department (/25):**

  Block size = 2^(32-25) = 128 addresses.
  - Network Address: 172.16.0.0/25
  - Valid Host IP Range: 172.16.0.1 to 172.16.0.126
  - Broadcast Address: 172.16.0.127
  - Usable Hosts: 2^7 - 2 = 126 usable addresses.

- **2. Sales Department (/26):** Starts at 172.16.0.128. Block size = 2^(32-26) = 64 addresses.
  - Network Address: 172.16.0.128/26
  - Valid Host IP Range: 172.16.0.129 to 172.16.0.190
  - Broadcast Address: 172.16.0.191
  - Usable Hosts: 2^6 - 2 = 62 usable addresses.

- **3. IT Department (/27):** Starts at 172.16.0.192.

  Block size = 2^(32-27) = 32 addresses.
  - Network Address: 172.16.0.192/27
  - Valid Host IP Range: 172.16.0.193 to 172.16.0.222
  - Broadcast Address: 172.16.0.223
  - Usable Hosts: 2^5 - 2 = 30 usable addresses.

- **4. Admin Department (/28):** Starts at 172.16.0.224.

  Block size = 2^(32-28) = 16 addresses.
  - Network Address: 172.16.0.224/28
  - Valid Host IP Range: 172.16.0.225 to 172.16.0.238
  - Broadcast Address: 172.16.0.239
  - Usable Hosts: 2^4 - 2 = 14 usable addresses.

Unused IP Address Range:

- **Allocated block range so far:** 172.16.0.0 to 172.16.0.239 (Total 240 addresses used in 172.16.0.0/24).
- **Remaining Unused Range in 172.16.0.0/16 block:** 172.16.0.240 to 172.16.255.255 (This includes 172.16.0.240/28 and 172.16.1.0 to 172.16.255.255).

Department	Subnet Mask	Network Address	Valid Host IP Range	Broadcast Address	Usable Hosts
HR	/25 (255.255.255.128)	172.16.0.0	172.16.0.1 – 172.16.0.126	172.16.0.127	126
Sales	/26 (255.255.255.192)	172.16.0.128	172.16.0.129 – 172.16.0.190	172.16.0.191	62
IT	/27 (255.255.255.224)	172.16.0.192	172.16.0.193 – 172.16.0.222	172.16.0.223	30
Admin	/28 (255.255.255.240)	172.16.0.224	172.16.0.225 – 172.16.0.238	172.16.0.239	14
Unused Space	N/A	172.16.0.240	172.16.0.240 – 172.16.255.255	N/A	65,296 addresses

Question 4:

Problem Analysis & Given Parameters:

- Given Base Block: 172.16.10.0/24
- Subnet Mask Used: /27 (255.255.255.224)

Subnetting Calculations:

- **1. Total Subnets Created:** Base prefix = /24, Subnet prefix = /27.
  Subnet bits borrowed (s) = 27 - 24 = 3 bits.
  - Total subnets = 2^s = 2^3 = 8 subnets.
- **2. Usable Hosts per Subnet:** Host bits remaining (h) = 32 - 27 = 5 bits.
  - Usable hosts per subnet = 2^h - 2 = 2^5 - 2 = 32 - 2 = 30 usable hosts.
- **3. Subnet Identification for IP 172.16.10.78:** Block size = 2^5 = 32.
  - Subnet intervals: Subnet 0 (0-31), Subnet 1 (32-63), Subnet 2 (64-95), Subnet 3 (96- 127).
  - Since 78 falls between 64 and 95:
    - Subnet to which 172.16.10.78 belongs: 172.16.10.64/27 (Subnet 2)
    - Network Address: 172.16.10.64
    - Broadcast Address: 172.16.10.95
    - Valid Host IP Range: 172.16.10.65 to 172.16.10.94
- **4. Verification of IP 172.16.10.95:** The IP 172.16.10.95 falls within the same subnet block (64-95). However, 172.16.10.95 is the BROADCAST ADDRESS for this subnet.
  - Therefore, while it belongs to the same subnet boundary, it is not a valid assignable host IP address.

Parameter	Value / Result
Base Network	172.16.10.0/24
Subnet Mask	/27 (255.255.255.224)
Total Subnets Created	8 subnets (2^3)
Usable Hosts per Subnet	30 hosts (2^5 - 2)
IP 172.16.10.78 Subnet Network Address	172.16.10.64
IP 172.16.10.78 Subnet Broadcast Address	172.16.10.95
Valid Host Range for 172.16.10.64/27	172.16.10.65 – 172.16.10.94
Is 172.16.10.95 in same Subnet?	Yes, but as the Broadcast Address (not a host IP)

Question 5:

Given IPv6 Address:

- Full IPv6 Address: 2001:0db8:0000:0000:0000:ff00:0042:8329
- Prefix Length: /64

Step-by-Step Solutions:

- **1. Address Compression (Shortest Form):** - Rule 1 (Omit leading zeros): 2001:db8:0:0:0:ff00:42:8329
  - Rule 2 (Zero compression using ::): Replace continuous sequence of zeros ":0:0:0:" with "::"
  - Compressed Address: 2001:db8::ff00:42:8329

• **2. Expansion Back to Full Notation:** - Identify missing 16-bit blocks: Original has 8 blocks, compressed has 5 blocks visible, so "::" expands to 3 blocks of "0000".
- Restore leading zeros to 4 hex digits per block.
- Expanded Address: 2001:0db8:0000:0000:0000:ff00:0042:8329

• **3. Network Prefix (/64):** - A /64 prefix means the first 64 bits (first 4 hextets) represent the network prefix.
- First 4 hextets: 2001:0db8:0000:0000
- Network Prefix: 2001:db8::/64 (or 2001:0db8:0000:0000::/64)

• **4. Total Usable / Possible Host Addresses:** - Interface ID bits remaining = 128 - 64 = 64 bits.
- Total possible host addresses = 2^64
- Calculation: 2^64 = 18,446,744,073,709,551,616 addresses (approx 18.44 quintillion addresses).

IPv6 Representation / Parameter	Value
Original IPv6 Address	2001:0db8:0000:0000:0000:ff00:0042:8329
Compressed Shortest Form	2001:db8::ff00:42:8329
Expanded Full Notation	2001:0db8:0000:0000:0000:ff00:0042:8329
Network Prefix (/64)	2001:db8::/64
Total Possible Host Addresses	2^64 = 18,446,744,073,709,551,616